

Swinburne University of Technology
School of Science, Computing, and Emerging Technologies

SWE40006 - Software Deployment and Evolution
Deployment Portfolio Task 2

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Submission date: [DO NOT FORGETTTTTTTTTTTTTTTTTTTTTTTTTTT]

Deployment Portfolio Task 2

1 Declared Target Level

All four subtasks 2.1, 2.2, 2.3, and 2.4 were attempted.

Table 1 below contain relevant links as proof of successful task execution.

Table 1: Publicly accessible URLs for live grading verification

Source	URL
Task 2.1	https://swe40006-task-2-1-atbmg9a7daapdnec.australiaeast-01.azurewebsites.net
Task 2.2	https://swe40006-task-2-2-f9emhph2gddzhker.australiaeast-01.azurewebsites.net/ (stopped)
Task 2.3	https://swe40006-task-2-3-hacthcdka0dub4ax.australiaeast-01.azurewebsites.net/ (stopped)

2 Execution Evidence

2.1 Task 2.1

For Task 2.1, I deployed a simple Node.js web application to Azure App Services, using Visual Studio Code and the Azure App Services extension.

First, I created the Node.js app consisting of the files `index.html`, `style.css`, `package.json` and `server.js`.



Figure 1: Structure of my simple Node.js application.

Then I ran my web application to test that it works correctly.

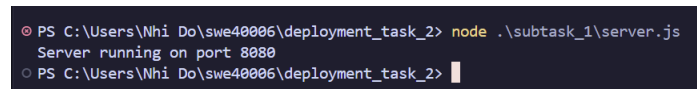


Figure 2: Running `npm start` on my Node.js application.

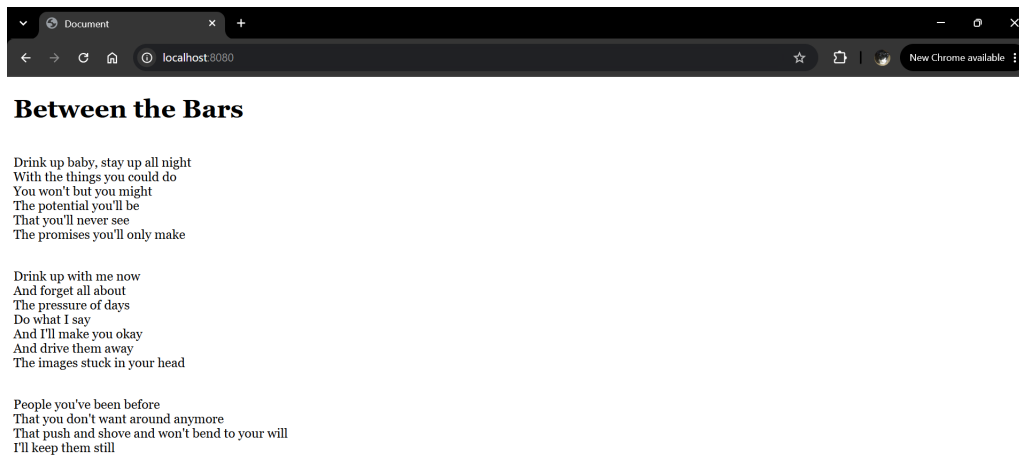


Figure 3: The web application runs locally on port 8080.

I then installed the Azure App Service on VS Code. The screenshot below shows the Azure panel once the extension has been successfully installed and I have logged in with my Azure account.

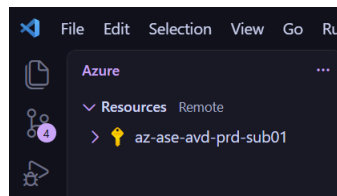


Figure 4: The Azure side panel on VS Code.

Next, I subscribed to an Azure for Students subscription on the Azure portal (portal.azure.com).

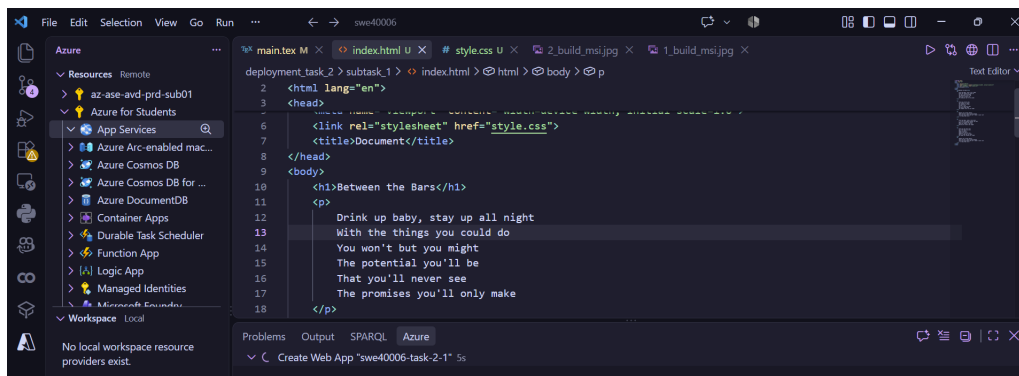


Figure 5: The Azure panel updated with my new Azure for Students subscription.

I then created a new Azure Web App service using the "Create Web App" command. I specified the region where I wanted my web app to be hosted which was Australia East, the application name `swe40006-task-2-1`, and the technology stack which was Node LTS 24.

As seen below, initial attempts to create the web app failed as my subscription was not registered with the `Microsoft.OperationalInsights` namespace. This was resolved by registering with the namespace on the Azure Portal.

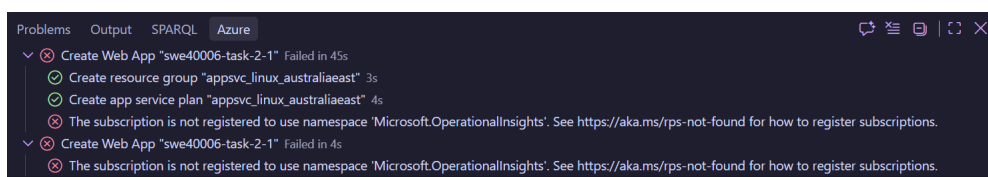


Figure 6: Failed attempts at creating a new Azure App Service instance.

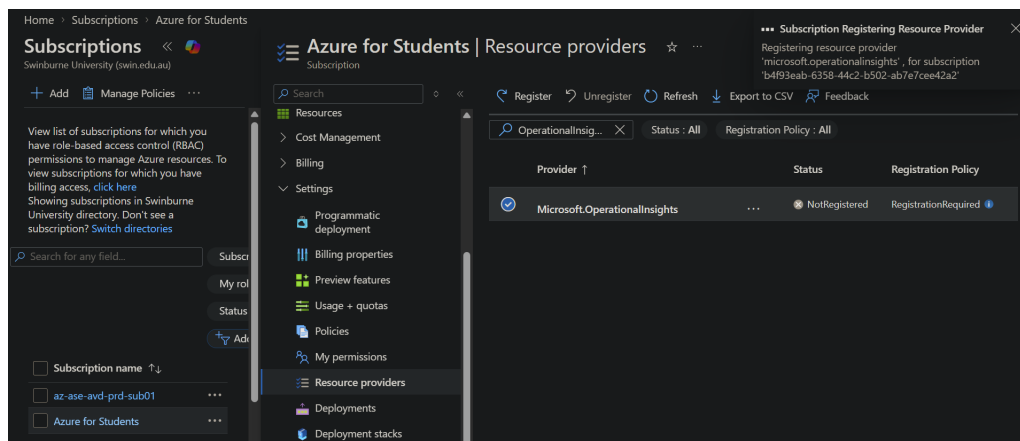


Figure 7: Registering with the Microsoft.OperationalInsights namespace on the Azure Portal.

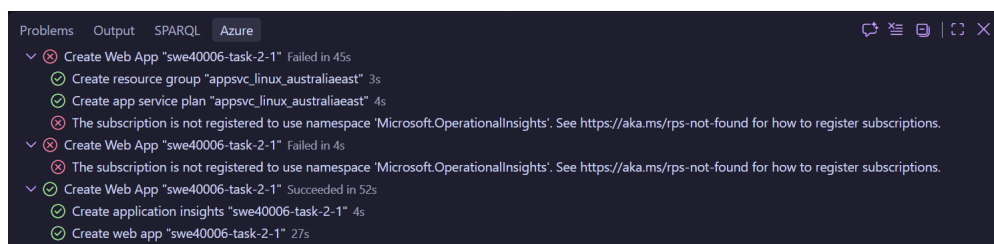


Figure 8: A new Azure App Service instance is successfully created.

The next step was to deploy my web application to the App Service instance.

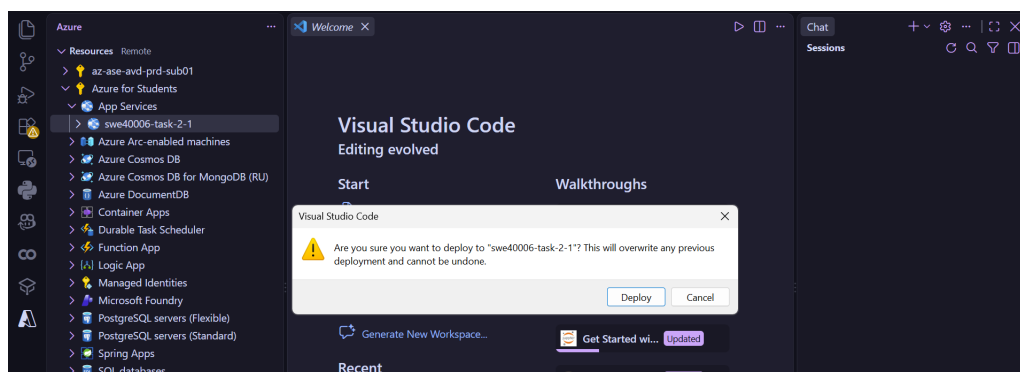


Figure 9: Deploying my web application to swe40006-task-2-1.

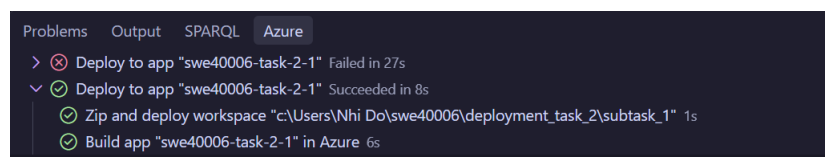
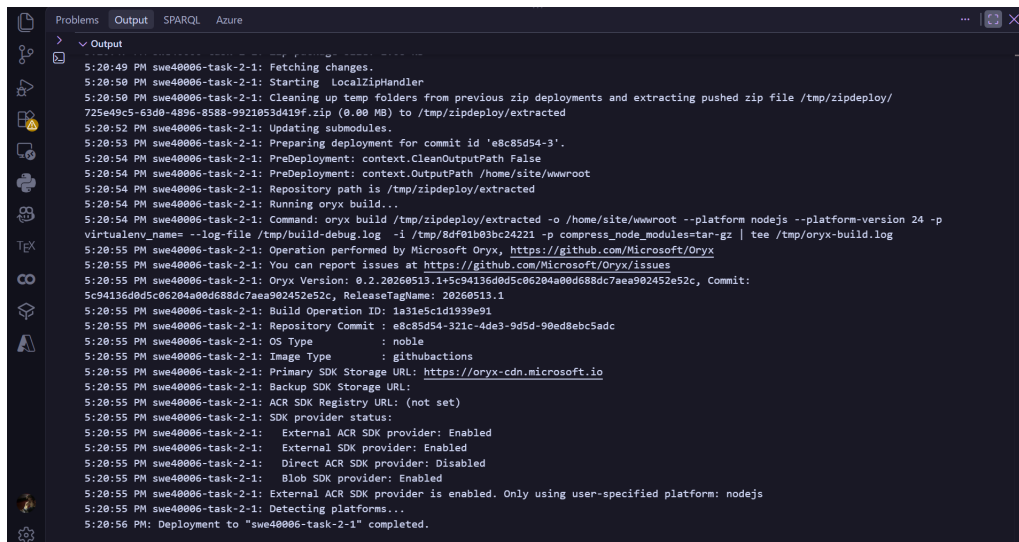


Figure 10: Deployment successful.



```

5:20:49 PM swe40006-task-2-1: Fetching changes.
5:20:50 PM swe40006-task-2-1: Starting LocalZipHandler
5:20:50 PM swe40006-task-2-1: Cleaning up temp folders from previous zip deployments and extracting pushed zip file /tmp/zipdeploy/725e49c5-63d0-4896-8588-9921053d419f.zip (0.00 MB) to /tmp/zipdeploy/extracted
5:20:52 PM swe40006-task-2-1: Updating submodules.
5:20:53 PM swe40006-task-2-1: Preparing deployment for commit id 'e8c85d54-3'.
5:20:54 PM swe40006-task-2-1: PreDeployment: context.CleanOutputPath False
5:20:54 PM swe40006-task-2-1: PreDeployment: context.OutputPath /home/site/wwwroot
5:20:54 PM swe40006-task-2-1: Repository path is /tmp/zipdeploy/extracted
5:20:54 PM swe40006-task-2-1: Running oryx build...
5:20:54 PM swe40006-task-2-1: Command: oryx build /tmp/zipdeploy/extracted -o /home/site/wwwroot --platform nodejs --platform-version 24 -p Virtualenv_name --log-file /tmp/build-debug.log -i /tmp/8d0f01b03bc24221 -p compress_node_modules=tar-gz | tee /tmp/oryx-build.log
5:20:55 PM swe40006-task-2-1: Operation performed by Microsoft Oryx, https://github.com/Microsoft/Oryx
5:20:55 PM swe40006-task-2-1: You can report issues at https://github.com/Microsoft/Oryx/issues
5:20:55 PM swe40006-task-2-1: Oryx Version: 0.2.20260513.145c94336d0d5c06204a00d688dc7aea902452e52c, Commit: 5c94136d0d5c06204a00d688dc7aea902452e52c, ReleaseTagName: 20260513.1
5:20:55 PM swe40006-task-2-1: Build Operation ID: 1a31e5c1d1939e91
5:20:55 PM swe40006-task-2-1: Repository Commit : e8c85d54-321c-4de3-9d5d-90ed8ebc5adc
5:20:55 PM swe40006-task-2-1: OS Type : noble
5:20:55 PM swe40006-task-2-1: Image Type : githubactions
5:20:55 PM swe40006-task-2-1: Primary SDK Storage URL: https://oryx-cdn.microsoft.io
5:20:55 PM swe40006-task-2-1: Backup SDK Storage URL:
5:20:55 PM swe40006-task-2-1: ACR SDK Registry URL: (not set)
5:20:55 PM swe40006-task-2-1: SDK provider status:
5:20:55 PM swe40006-task-2-1: External ACR SDK provider: Enabled
5:20:55 PM swe40006-task-2-1: External SDK provider: Enabled
5:20:55 PM swe40006-task-2-1: Direct SDK provider: Disabled
5:20:55 PM swe40006-task-2-1: Blob SDK provider: Enabled
5:20:55 PM swe40006-task-2-1: External ACR SDK provider is enabled. Only using user-specified platform: nodejs
5:20:55 PM swe40006-task-2-1: Detecting platforms...
5:20:56 PM: Deployment to "swe40006-task-2-1" completed.

```

Figure 11: Build output.

The application is live and publicly accessible at <https://swe40006-task-2-1-atbmg9a7daapdnec.australiaeast-01.azurewebsites.net>.

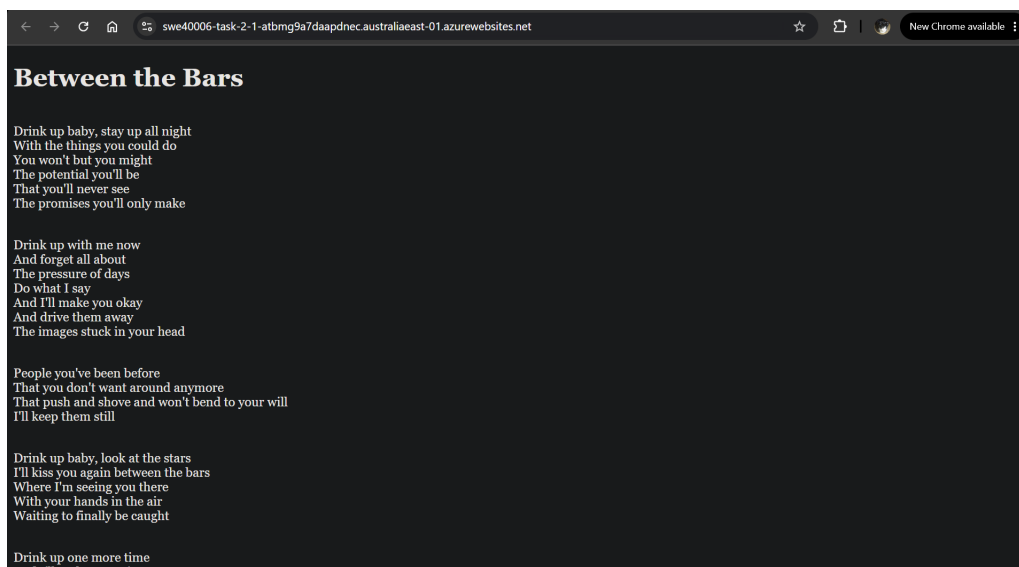


Figure 12: The application is deployed and running live.

2.2 Task 2.2

For task 2.2, I deployed an ASP.NET Razor Pages application with Azure App Services, also using Visual Studio Code and the Azure App Service extension.

First, I created a new Razor Pages project using the .NET CLI. Once the project has been created, I ran `dotnet publish` to publish the application.

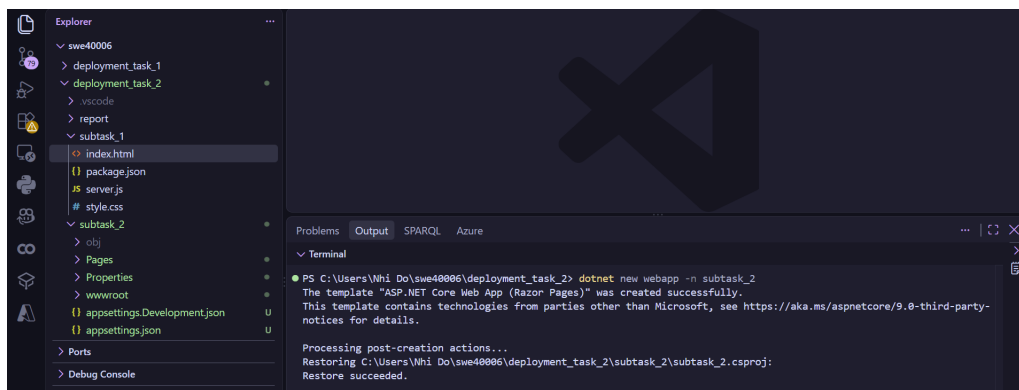


Figure 13: Creating a new Razor Pages project via the CLI.

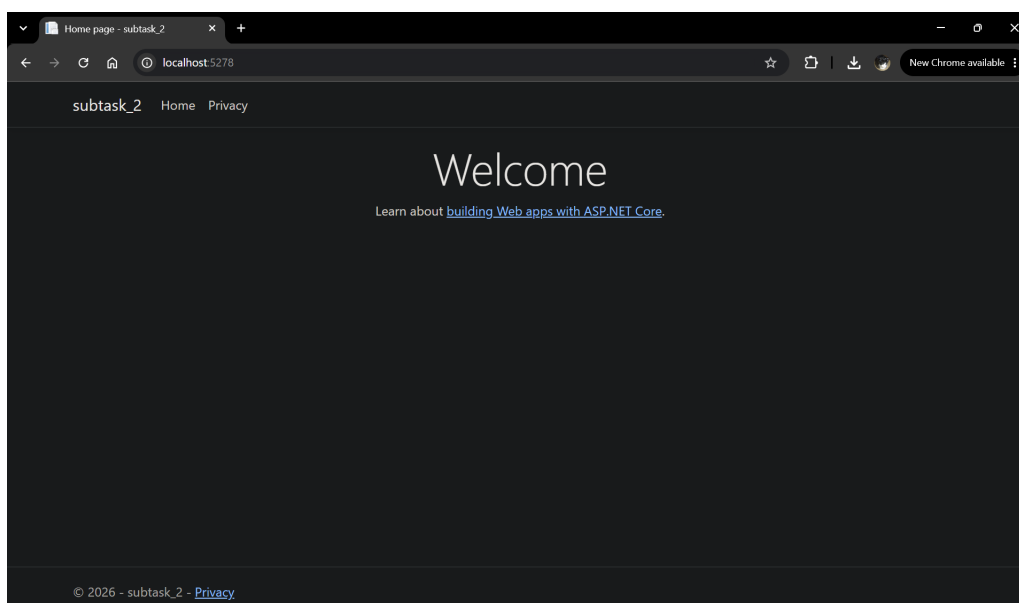


Figure 14: The application runs locally on port 5278.

Similar to Task 2.1, I created a new Web App Service instance using the Azure App Service extension. The new instance is named **swe40006-task-2-2**, hosted in the Australia East region, and using the .NET 8.0 runtime.

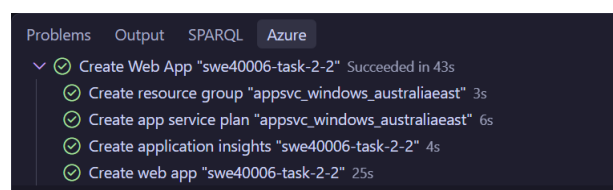


Figure 15: A new Web App Service instance is successfully created.

The application is then deployed into this new instance. It is hosted at <https://swe40006-task-2-2-f9emhph2gddzhker.australiaeast-01.azurewebsites.net/>.

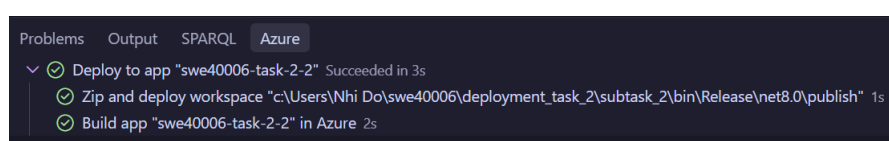
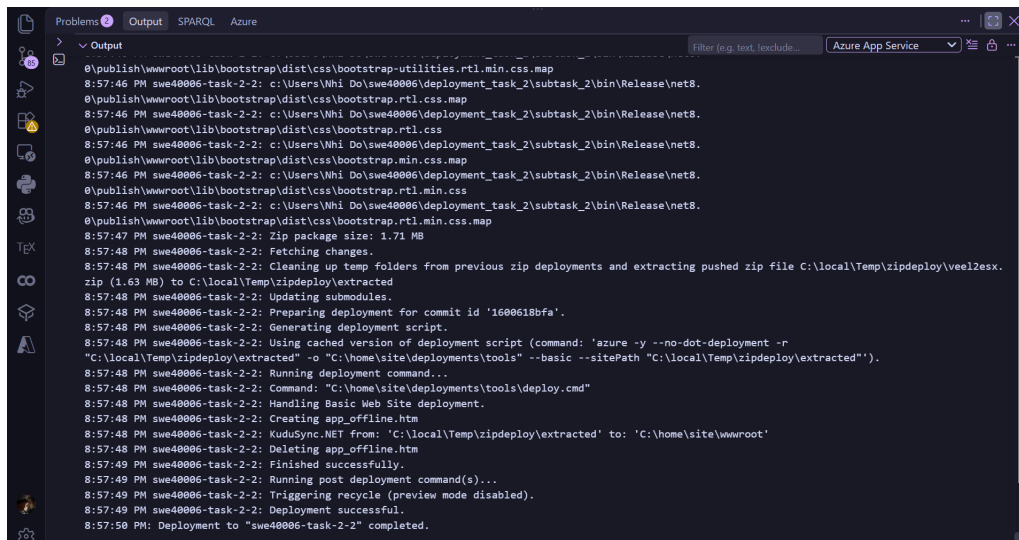


Figure 16: Successful deployment.



```

0:\publish\wwwroot\lib\bootstrap\dist\css\bootstrap-utilities.rtl.min.css.map
8:57:46 PM swe40006-task-2-2: c:\Users\Nhi Do\swe40006\deployment_task_2\subtask_2\bin\Release\net8.
0:\publish\wwwroot\lib\bootstrap\dist\css\bootstrap.rtl.css.map
8:57:46 PM swe40006-task-2-2: c:\Users\Nhi Do\swe40006\deployment_task_2\subtask_2\bin\Release\net8.
0:\publish\wwwroot\lib\bootstrap\dist\css\bootstrap.rtl.css
8:57:46 PM swe40006-task-2-2: c:\Users\Nhi Do\swe40006\deployment_task_2\subtask_2\bin\Release\net8.
0:\publish\wwwroot\lib\bootstrap\dist\css\bootstrap.min.css.map
8:57:46 PM swe40006-task-2-2: c:\Users\Nhi Do\swe40006\deployment_task_2\subtask_2\bin\Release\net8.
0:\publish\wwwroot\lib\bootstrap\dist\css\bootstrap.rtl.min.css
8:57:46 PM swe40006-task-2-2: c:\Users\Nhi Do\swe40006\deployment_task_2\subtask_2\bin\Release\net8.
0:\publish\wwwroot\lib\bootstrap\dist\css\bootstrap.rtl.min.css.map
8:57:47 PM swe40006-task-2-2: Zip package size: 1.71 MB
8:57:48 PM swe40006-task-2-2: Fetching changes.
8:57:48 PM swe40006-task-2-2: Cleaning up temp folders from previous zip deployments and extracting pushed zip file C:\local\Temp\zipdeploy\veel2esx.
zip (1.63 MB) to C:\local\Temp\zipdeploy\extracted
8:57:48 PM swe40006-task-2-2: Updating submodules.
8:57:48 PM swe40006-task-2-2: Preparing deployment for commit id '1600618bfa'.
8:57:48 PM swe40006-task-2-2: Generating deployment script.
8:57:48 PM swe40006-task-2-2: Using cached version of deployment script (command: 'azure -y --no-dot-deployment -r
"C:\local\Temp\zipdeploy\extracted" -o "C:\home\site\deployments\tools" --basic --sitePath "C:\local\Temp\zipdeploy\extracted"').
8:57:48 PM swe40006-task-2-2: Running deployment command...
8:57:48 PM swe40006-task-2-2: Command: "C:\home\site\deployments\tools\deploy.cmd"
8:57:48 PM swe40006-task-2-2: Handling Basic Web Site deployment.
8:57:48 PM swe40006-task-2-2: Creating app_offline.htm
8:57:48 PM swe40006-task-2-2: KuduSync.NET from: 'C:\local\Temp\zipdeploy\extracted' to: 'C:\home\site\wwwroot'
8:57:48 PM swe40006-task-2-2: Deleting app_offline.htm
8:57:49 PM swe40006-task-2-2: Finished successfully.
8:57:49 PM swe40006-task-2-2: Running post deployment command(s)...
8:57:49 PM swe40006-task-2-2: Triggering recycle (preview mode disabled).
8:57:49 PM swe40006-task-2-2: Deployment successful.
8:57:50 PM: Deployment to "swe40006-task-2-2" completed.

```

Figure 17: Logs showing that deployment was successful.

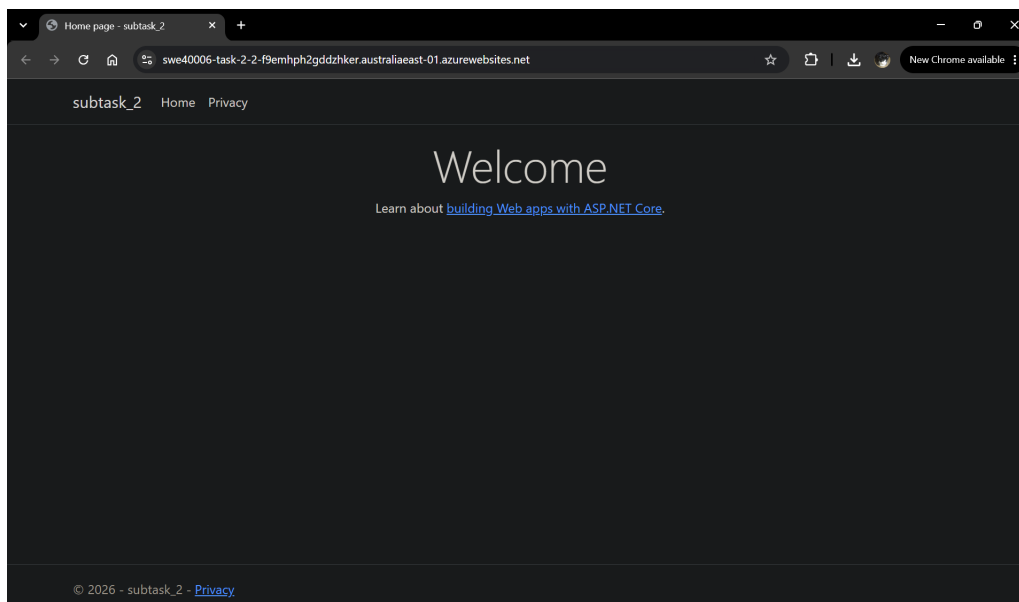


Figure 18: The web application up and running live.

Even though this App Service instance is running on a free tier, it is still best practice to manage cloud resources responsibly to avoid unintended charges. To demonstrate this, I stopped the instance using the Azure Portal.

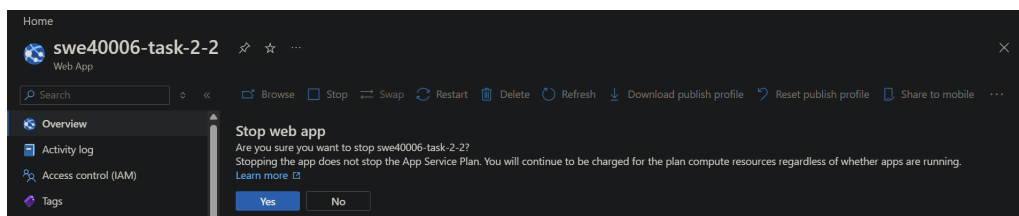


Figure 19: Stopping the instance.

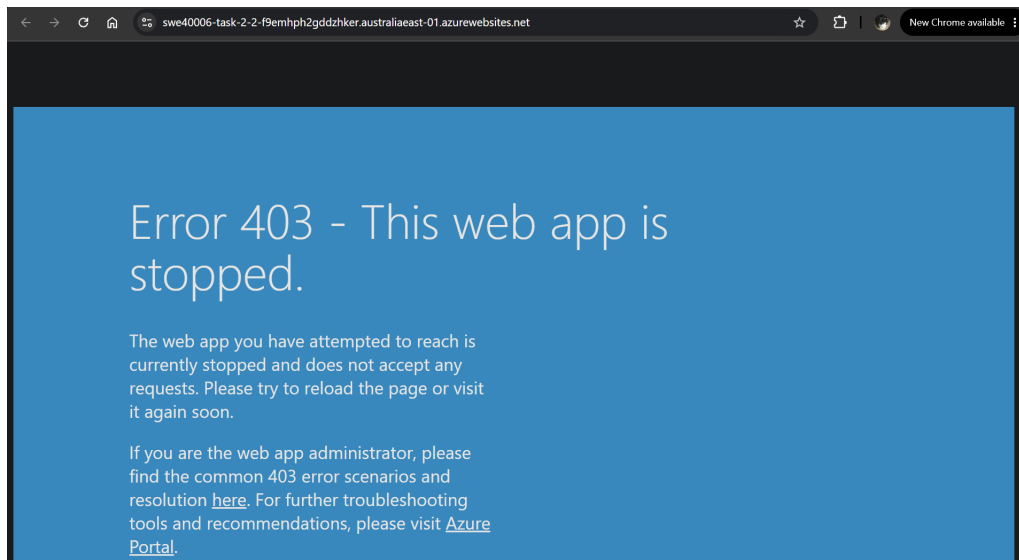


Figure 20: The web app has been stopped.

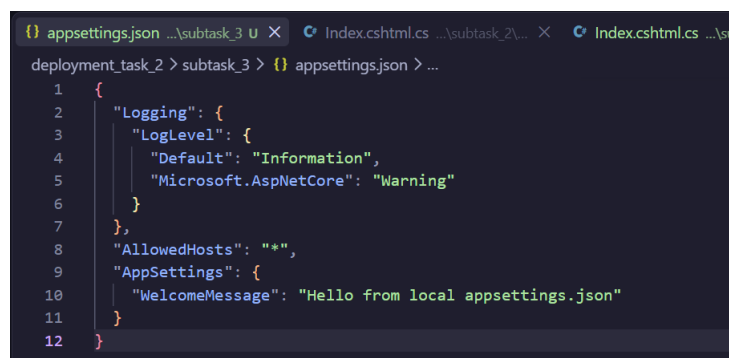
2.3 Task 2.3

For task 2.3, I deployed a copy of the Razor Pages application created in the previous task.

2.3.1 Configuration Management

To demonstrate configuration abstraction, I defined a setting in `appsettings.json` and read its value at runtime via `IConfiguration`, rather than hardcoding it directly in the application code. This value was then overridden using an Azure App Service Environment Variable in the Azure Portal, allowing the deployed application to resolve a different value. In a real scenario involving sensitive data, the value in `appsettings.json` would be left blank, and the real value stored locally using `user-secrets`. As the configuration value here is non-sensitive and only for demonstration purposes, I kept it in `appsettings.json` for easy testing.

First, I added the new setting `WelcomeMessage` in `appsettings.json`.

Figure 21: Adding a new setting in `appsettings.json`.

I then tested the application locally. The configuration value showed up as expected.

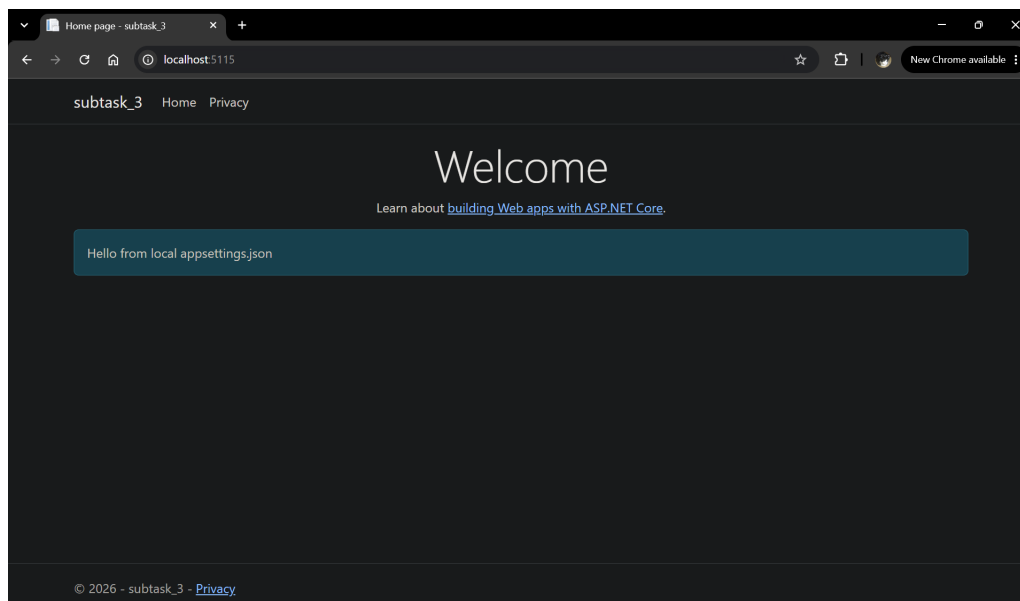


Figure 22: Testing that the app runs fine locally.

I then deployed the app with the Azure App Service extension similar to what I did in task 2.2.

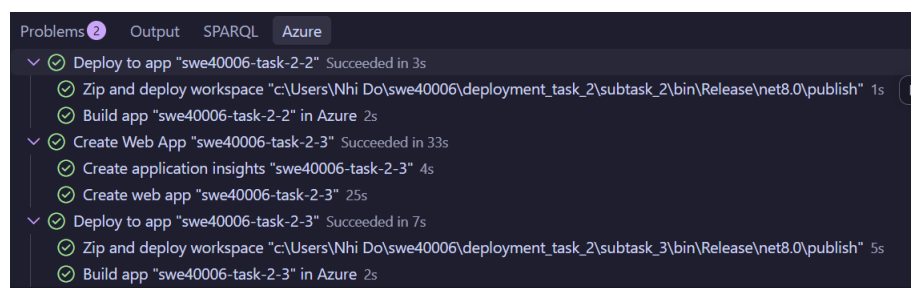
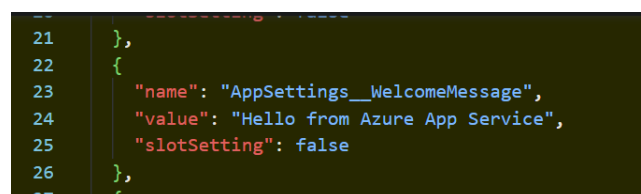


Figure 23: Creating a new Web Service instance and deploying my app.

In the Azure Portal, I added a new environment variable to the newly created `swe40006-task-2-3` App Service. The value here is different from the value that was used locally ("Hello from Azure Web Service" vs. "Hello from local appsettings.json") to demonstrate the overriding mechanism.

Figure 24: Adding a new environment variable `AppSettings__WelcomeMessage` in the Azure Portal.

The overridden value set to the environment variable showed up on the deployed website.

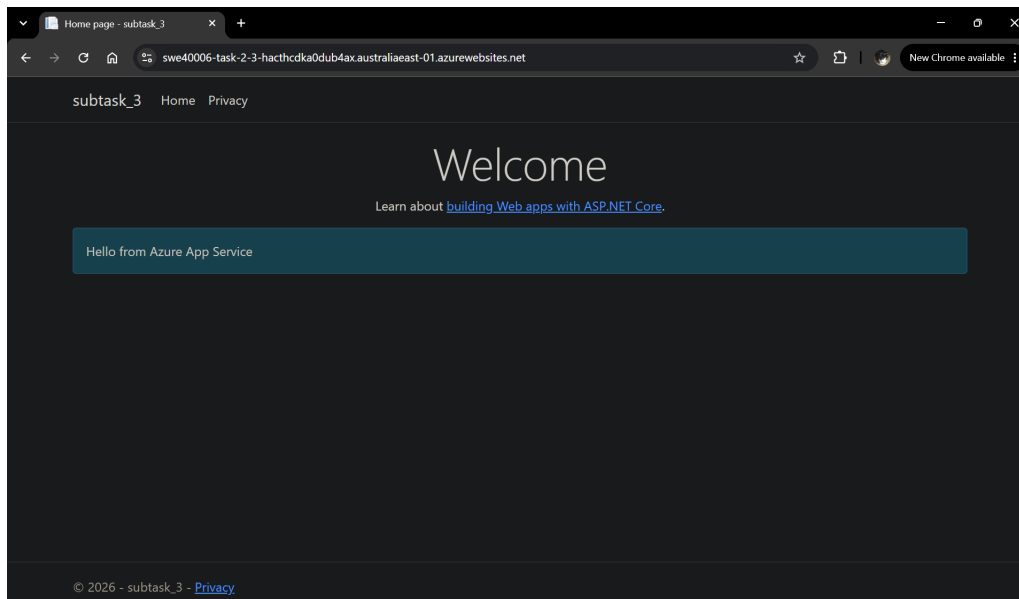


Figure 25: The site showing the new value.

2.3.2 Diagnostics and Logging

I also configured a active health check on this `swe40006-task-2-3` instance. First, I modified the source code to the the health check endpoint.

```
26 app.MapHealthChecks("/health");  
27 app.Run();
```

```
2 // Add services to the container.  
3 builder.Services.AddRazorPages();  
4 builder.Services.AddHealthChecks();  
5  
6
```

Figure 26: Adding a new health check endpoint to the app.

Then, once on the Azure Portal, I realised that the health check wasn't available on the free tier. This was resolved by scaling the instance to a Basic plan.

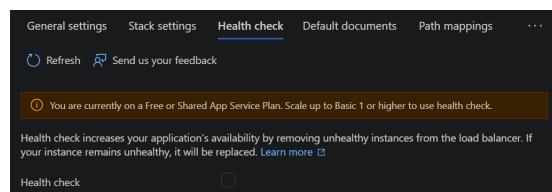


Figure 27: The health check is not available for free-tier instances.

Once that was resolved, I could configure the health check on the portal by adding the path to the health check endpoint, which was `/health`.

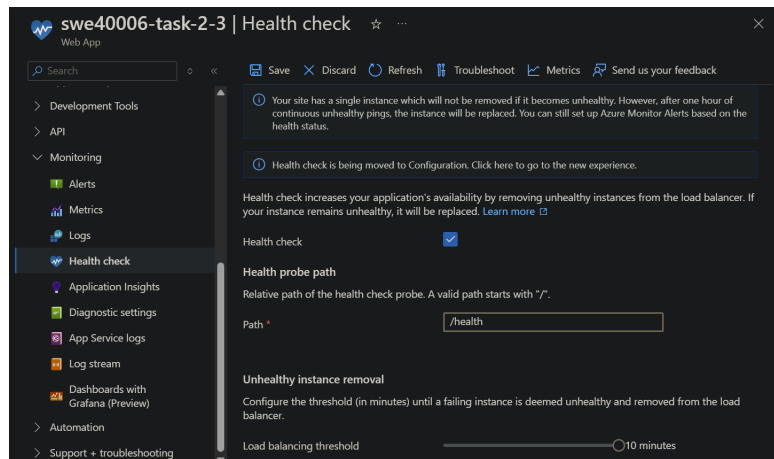


Figure 28: Setting up the active health check endpoint on Azure Portal.

Going to the endpoint at <https://swe40006-task-2-3-hacthcdka0dub4ax.australiaeast-01.azurewebsites.net/health> verified that the instance was healthy. I then stopped the instance to ensure that resources weren't being wasted.

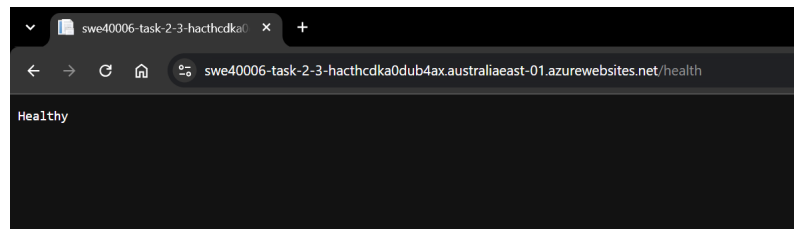


Figure 29: Verifying that active health checking is working.